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To cite this article: Hongming Shi 2025 *J. Phys.: Conf. Ser.* **2977** 012018

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Design and aerodynamic analysis of a samara-inspired unmanned aerial vehicle

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Abstract. This paper presents the design and aerodynamic analysis of a novel unmanned aerial vehicle (UAV) inspired by the samara seed's natural flight mechanics. The UAV is designed to emulate the self-stabilizing spin behavior of the winged seed during descent, offering an innovative approach to UAV swarm applications. The aircraft incorporates a spinning wing, a load-bearing fuselage, and integrated propulsion and control systems, optimized for both military and civilian use. Using computational fluid dynamics (CFD) simulations, the aerodynamic performance of the UAV was evaluated under various operational conditions, focusing on lift generation, stability, and control dynamics. The results demonstrate that the samara-inspired design achieves stable spin flight and adequate lift to carry a payload while maintaining lateral stability. Potential applications for this design include precision aerial delivery, reconnaissance, and environmental monitoring.

1. Introduction

Conventional UAV designs, particularly fixed-wing and rotary-wing models, face limitations in low-speed stability, vertical descent, and hovering. Fixed-wing UAVs provide excellent aerodynamic efficiency but lack the maneuverability necessary for precise, localized tasks, such as those required in search-and-rescue operations or environmental sampling. Rotary-wing UAVs, such as quadcopters, offer hovering capabilities but often suffer from lower efficiency and shorter flight times [1–3].

In response to these limitations, researchers have increasingly turned to bio-inspired designs, particularly those that emulate the natural flight characteristics of winged seeds, such as samaras. Samara seeds, found in trees like maples, spin as they fall, generating lift and prolonging their descent [4–5]. This natural mechanism of autorotation provides inherent stability and has the potential to enhance the capabilities of UAVs, particularly in swarm applications where stability and extended flight times are critical [6–7].

Several studies have explored the application of samara-inspired designs in UAVs. For instance, Win et al. [8] developed a UAV prototype that achieved stable spin flight by leveraging the aerodynamic properties of a single rotating wing. This work demonstrated the feasibility of using natural flight mechanisms to enhance UAV performance. Soh et al. [9] and Thomas et al. [10] introduced propulsion systems to these designs, which allowed for greater maneuverability and control during flight. These efforts have significantly advanced the understanding of bio-inspired UAVs, particularly in terms of aerodynamic performance.

Despite these advances, challenges remain. Most existing designs focus on basic aerodynamic performance, with limited attention to control system integration or payload capacity. Moreover, the scalability of samara-inspired UAVs for swarm applications has not been thoroughly explored. Other

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researchers, such as Lee et al. [11] and Win et al. [12], have begun investigating the role of advanced control systems in samara-like UAVs, but further research is needed to fully realize their potential.

This study seeks to address these gaps by presenting a comprehensive design and analysis of a samara-inspired UAV, focusing on both aerodynamic performance and control system integration. By combining bio-inspired design principles with modern UAV technologies, we aim to create a UAV that offers superior stability, extended flight times, and precise control, making it suitable for a wide range of applications.

2. Design of the samara-inspired UAV

2.1 Conceptual design and structural layout

The design of the samara-inspired UAV draws directly from the unique aerodynamic properties of samara seeds, as shown in Figure 1(a), which generate lift through autorotation as they descend. A comparative analysis of samara seed biomechanics reveals several critical principles that inform this UAV's design. First, the asymmetric wing structure, typically observed in samara seeds, creates an aerodynamic imbalance that facilitates rotational motion, generating lift through a continuous low-pressure vortex along the wing's upper surface. This UAV design incorporates an elongated, single-wing configuration, optimized for a controlled descent through steady autorotation. The single rotating wing, spanning 400 mm with a chord length of 80 mm, mimics the seed's ability to convert vertical descent into rotational kinetic energy, stabilizing flight and minimizing power consumption.

The wing is modeled after the flat, elongated structure of the samara seed, which uses autorotation to maintain a slow, controlled descent. Inspired by the thin, flexible profile of samara wings, the UAV utilizes a Clark Y airfoil with a maximum thickness of 11.7% of the chord length. This design, similar to the seed's wing, balances lift and drag to sustain a low descent rate, thereby enhancing stability. Additionally, the Clark Y airfoil's high lift-to-drag ratio and favorable stall characteristics ensure that the UAV remains controllable even at low speeds, a key advantage for tasks requiring precision. Another essential design principle derived from samara biomechanics is mass distribution, which is crucial for maintaining rotational stability. The wing's span is 400 mm, while the chord length measures 80 mm, resulting in a total wing area of 0.032 m². This size was chosen after multiple iterations and calculations to balance lift generation with structural simplicity. The Clark Y airfoil, well-suited for low-speed flight, was selected for its high lift-to-drag ratio and favorable stall characteristics, as shown in Figure 1(b). The Clark Y airfoil has a maximum thickness of 11.7% of the chord length, providing the strength needed to withstand aerodynamic loads while maintaining flexibility for autorotation.

The propulsion system further integrates bio-inspired principles, using a single brushless motor mounted at the wingtip. This motor provides the initial thrust required to initiate autorotation, after which minimal power is needed to sustain rotational flight, emulating the energy efficiency of samara seeds that descend with no active propulsion. By only supplementing with power when necessary, this UAV design capitalizes on natural aerodynamic stability to achieve a longer flight duration with reduced energy consumption. This motor, a TOKA 2004-1700KV model, is selected for its balance between power output and lightweight construction. With a KV rating of 1700 and a maximum power output of 418.1 W, the motor provides sufficient torque to spin the wing at speeds up to 10 rad/s. The propeller, a 5-inch carbon fiber TAROT 5045, ensures efficient thrust generation while minimizing drag, as shown in Figure 1(c). This design enables the UAV to maintain a rotational speed necessary for stable flight, generating the lift needed to support the payload.

2.2 Wing design and aerodynamic considerations

The wing design plays a pivotal role in the UAV's ability to generate lift through autorotation. The rectangular planform, combined with the Clark Y airfoil, was chosen after a thorough evaluation of several airfoil profiles. The Clark Y airfoil is particularly suitable due to its performance in low-speed regimes, such as those encountered during descent. At an angle of attack of 6°, the airfoil achieves a

lift coefficient of 1.2 and a drag coefficient of 0.02. The maximum lift-to-drag ratio is 60, which ensures that the UAV can sustain its flight for extended periods without excessive energy consumption.

In terms of mass distribution, the design ensures that the center of mass is aligned with the rotation axis, which minimizes instability during flight. The lightweight construction of the wing and fuselage also reduces the overall moment of inertia, allowing the UAV to rotate smoothly and maintain lateral stability.

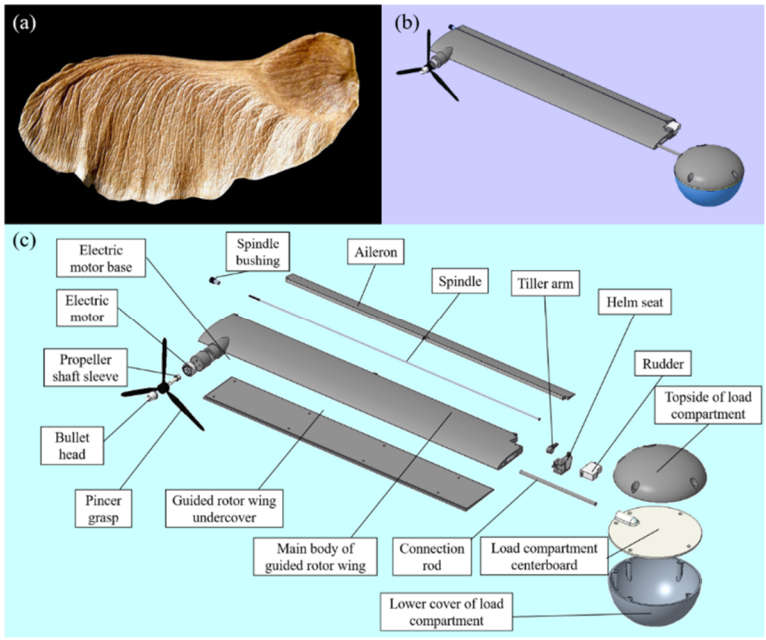


Figure 1. Global design of the samara-inspired UAV. (a) The structure of samara. (b) General view of samara-inspired UAV. (c) Exploded view of samara-inspired UAV.

It is also seen that the specimen SJ1 has the largest deformation, and the maximum deflection of the specimen is up to 16 mm, which shows the better plastic deformation ability of the slab. It is also seen that the specimen SJ1 has the largest deformation, and the maximum deflection of the specimen is up to 16 mm, which shows the better plastic deformation ability of the slab. It is also seen that the specimen SJ1 has the largest deformation, and the maximum deflection of the specimen is up to 16 mm, which shows the better plastic deformation ability of the slab.

2.3 Control systems and propulsion

The control system is designed to adjust the angle of attack dynamically, thus enabling the UAV to control its descent rate, maintain stability, and alter its horizontal trajectory. Two digital servomotors are mounted at the base of the wing, controlling the pitch angle. These servos, with a torque of 1.8 kg/cm and a response frequency of 144 Hz, provide precise control over the wing's angle of attack, allowing the UAV to react swiftly to changes in flight conditions.

The servos are integrated with the flight controller, which processes data from the UAV's onboard sensors, including accelerometers, gyroscopes, and an altimeter. The flight controller continuously monitors the UAV's orientation and flight parameters, adjusting the wing's angle of attack in real time to ensure stable flight. This closed-loop control system enables the UAV to adapt to environmental changes, such as gusts of wind, while maintaining its intended flight path.

The propulsion system is powered by an 11.1 V, 2200 mAh Li-Po battery, providing enough energy for approximately 20 minutes of continuous flight under normal conditions. The motor operates efficiently at rotational speeds between 5000 and 10,000 RPM, generating a maximum thrust of 1.8 N. During powered flight, the motor provides additional thrust to extend the UAV's range, while in autorotation mode, it operates at reduced power to conserve battery life.

3. Aerodynamic simulation and analysis

3.1 CFD methodology

The aerodynamic performance of the samara-inspired UAV was evaluated using computational fluid dynamics (CFD) simulations in ANSYS Fluent. To enhance the realism of these simulations, additional tests were conducted under complex flight environments that emulate practical application conditions. Instead of relying solely on idealized, free-stream conditions, we incorporated non-standard atmospheric variables, including variable air density, temperature gradients, and simulated turbulence, to better reflect the UAV's performance in real-world scenarios. This approach provides a comprehensive understanding of the UAV's stability, lift, and energy consumption across diverse environmental conditions. The primary objectives were to analyze the lift generated by the spinning wing, examine the lateral stability of the UAV during descent, and assess the drag and torque forces acting on the system.

One key factor considered was air density variation, which can significantly impact the UAV's lift and drag. In high-altitude scenarios, where air density decreases, the UAV's lift generation capacity is naturally reduced. Conversely, low-altitude, high-density conditions, such as near sea level, enhance lift but increase drag, potentially affecting the UAV's energy efficiency. By simulating a range of altitudes, from 0 to 3000 meters above sea level, the CFD analysis revealed how the UAV's flight characteristics adapt to different air densities, providing insight into operational efficiency across a variety of altitudes. Temperature gradients were also introduced, simulating conditions ranging from cold environments (0°C) to high temperatures (40°C). These temperature shifts impact air viscosity and density, subtly altering the lift-to-drag ratio. The CFD results showed that, under high-temperature conditions, the UAV required a slightly higher rotational speed to achieve the same lift as in cooler environments, underscoring the importance of temperature-aware control adjustments in operational planning. Turbulence was a further variable added to the CFD simulations to model sudden wind gusts and unsteady airflow.

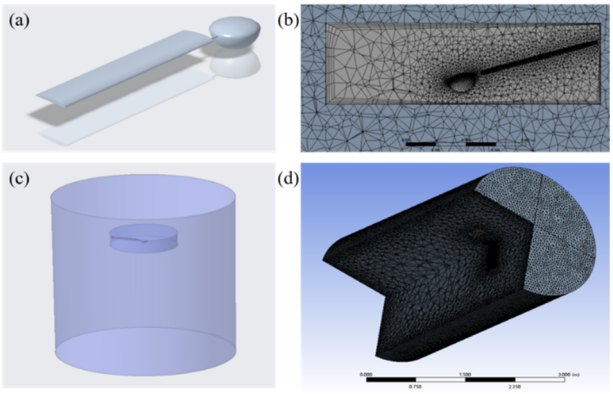


Figure 2. CFD finite element simulation modeling. (a) Simplified model of the samara-inspired UAV. (b) Grid effect diagram of samara-inspired UAV. (c) Dynamic lattice regions and outflow fields. (d) Unstructured grid based on flow field structure.

The CFD simulations used a 3D model of the UAV, which was developed in CAD software and imported into ANSYS Fluent for analysis, as shown in Figures 2(a-b). The simulation environment was configured to simulate free-stream conditions typical of flight at altitudes between 50 and 100 m. The Reynolds-Averaged Navier-Stokes (RANS) equations were used to model airflow, with a k-epsilon turbulence model to capture the effects of rotational flow around the spinning wing.

A cylindrical control volume was created around the UAV to represent the surrounding air, with boundary conditions set to ensure that airflow remained uniform and undisturbed at the domain's edges, as shown in Figures 2(c-d). The free-stream velocity was set to 5 m/s, representing typical operating conditions in light wind. The air density was set at 1.225 kg/m^3, corresponding to standard atmospheric conditions.

The wing's rotational speed was varied between 4 and 10 rad/s, with the angle of attack adjusted between 2° and 16° to assess the UAV's performance under different flight conditions. The UAV's drag, lift, and torque coefficients were monitored throughout the simulations to provide detailed insight into its aerodynamic behavior.

3.2 Simulation setup and parameters

The UAV's wing was modeled using an unstructured mesh to capture the fine details around the leading and trailing edges of the airfoil. This mesh was refined at critical regions, such as the wing's tips and around the fuselage, to ensure accurate predictions of flow separation and vortex shedding. The mesh size near the wing was reduced to 0.002 m to resolve the boundary layer accurately, while the overall mesh consisted of approximately 3 million elements.

The wing rotation rate was simulated at three distinct speeds - 4, 6, 8, and 10 rad/s - to evaluate its performance under varying conditions. At each rotational speed, the wing's angle of attack was varied from 2° to 16° in 2° increments. This range of angles was chosen to ensure that both low-lift and high-lift configurations were analyzed, as well as potential stall conditions.

The simulation was conducted as a transient analysis to capture the unsteady nature of the flow around the spinning wing. Each simulation ran for 0.5 seconds of real time, which corresponds to approximately 30 complete rotations of the wing at the lowest speed. The time step was set to 0.001 seconds to ensure sufficient resolution of the wing's rotational motion.

3.3 Results: lift, drag, and torque analysis

The CFD simulations revealed several key insights into the aerodynamic performance of the UAV's wing. In Figure 3(a), which shows the lifting force as a function of the facing angle for various rotational speeds, additional data points have been added at higher rotational speeds (up to 18 rad/s). These data points provide a more comprehensive view of the lift dynamics across a wider range of operational conditions, highlighting how lift generation increases significantly with both the angle of attack and the rotational speed. At lower rotational speeds (4-8 rad/s), the lift generated is relatively modest, showing gradual increases as the facing angle rises. This trend indicates that at low RPMs, the UAV's lifting capacity is limited, and achieving substantial lift would require a high angle of attack, which may lead to increased drag and potential instability. At a rotational speed of 10 rad/s and an angle of attack of 6°, the UAV generated a lift force of 4.2 N, sufficient to support a payload of 200 g, as shown in Figure 3(a). The lift coefficient reached a maximum of 1.3, while the drag coefficient remained low at 0.03, resulting in a high lift-to-drag ratio of 43.3. However, as the rotational speed increases beyond 10 rad/s, the lift generated at each facing angle shows a marked increase. For instance, at 12 rad/s and a facing angle of 10°, the lift force reaches approximately 6 N, while at 18 rad/s, the lift force at the same angle nearly doubles to 12 N. This relationship underscores a critical design insight: by operating at higher rotational speeds, the UAV can achieve significant lift without requiring excessively high facing angles, which reduces the aerodynamic drag typically associated with large angles of attack. The data also reveal that at extreme angles, close to 16°, lift generation begins to plateau or increase at a slower rate even with higher RPMs. This plateau suggests the onset of aerodynamic limits, where further increases in the angle of attack yield diminishing returns in lift due to flow separation effects at the wing's trailing edge. These results confirm that the wing design is capable of maintaining stable flight and achieving the necessary lift for operational tasks.

The torque analysis showed that the rotational motion remained stable across all tested angles of attack. As the facing angle increases, the resistance torque initially remains stable but then rises sharply after 10°, reaching a maximum at 16°. At a speed of 10 rad/s, the torque generated by aerodynamic forces acting on the wing reached a peak of 0.35 N·m, as shown in Figure 3(b). This value was well within the operating limits of the UAV's motor, confirming that the system can maintain autorotation without excessive energy consumption. These results validate the UAV's design strategy of balancing rotational speed and angle of attack to optimize lift without incurring excessive

resistance torque. This balance is essential for sustaining flight efficiency, as higher torque demands would lead to greater energy consumption.

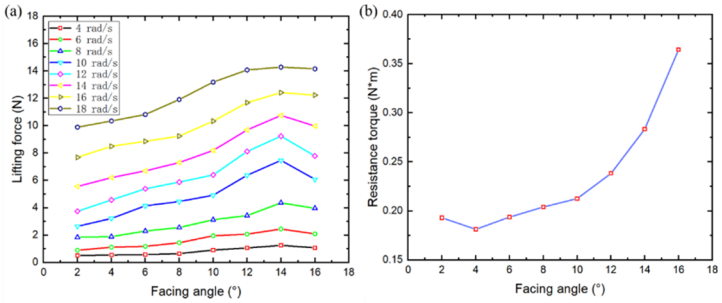


Figure 3. Simulation results analysis. (a) Variation curves of lift as a function of the orientation angle of the guided rotor wing for different rotation speeds. (b) Variation curve of drag moment with the angle of orientation of the guided rotor wing.

4. Control mechanisms

4.1 Design of the control system

The control system for the samara-inspired UAV is designed to ensure stable spin flight while providing the capability to adjust descent rate and horizontal movement. Unlike traditional UAVs that rely on multiple rotors for control, the samara-inspired UAV uses two servomotors to dynamically adjust the wing's angle of attack. These servos allow the UAV to control its lift and drag, effectively managing its flight path and stability.

The servos are connected to the flight controller, which continuously monitors the UAV's orientation using data from onboard sensors. The accelerometer and gyroscope provide real-time feedback on the UAV's rotational speed and angular velocity, while the altimeter measures the UAV's altitude. This data is processed by the flight controller's PID (Proportional-Integral-Derivative) control algorithm, which adjusts the servos to maintain stable flight.

4.2 Spin flight stabilization

Maintaining stability during spin flight is a key challenge for the UAV. The spinning wing generates both lift and centrifugal forces, which must be carefully balanced to avoid instability. To achieve this balance, the servomotors adjust the wing's pitch in response to changes in the UAV's flight dynamics. For example, if the UAV's descent rate increases too rapidly, the servos will increase the angle of attack to generate more lift and slow the descent.

In addition to controlling descent, the servos also allow for lateral maneuvering. By adjusting the pitch of one side of the wing more than the other, the UAV can generate a yawing moment, enabling it to turn during flight. This level of control makes the UAV suitable for a variety of applications, including precision landing and aerial delivery.

4.3 Integration of sensors and actuators

The UAV's control system integrates several key components to ensure precise control during flight. The accelerometers and gyroscopes provide data on the UAV's rotational motion, while the altimeter monitors altitude changes. These sensors feed real-time data into the flight controller, which uses this information to adjust the wing's pitch and rotational speed. To enhance robustness, additional response strategies were developed to handle potential sensor failure or data loss, ensuring continuous operation even under adverse conditions. In the event of sensor failure, a hierarchical fallback system activates. For instance, if the primary gyroscope fails, the control system automatically switches to backup gyroscopes or relies on accelerometer data to approximate the UAV's orientation. This redundancy allows the UAV to maintain stability until repairs can be made. The servomotors, capable of high-

frequency adjustments, allow the UAV to respond quickly to environmental changes, such as wind gusts or obstacles.

The integration of these components ensures that the UAV can maintain stable flight under a wide range of conditions. The servomotors' ability to adjust the wing's angle of attack in real time enables the UAV to maintain its desired flight path, even in turbulent environments.

5. Results and discussion

The CFD simulations and control system analysis provided valuable insights into the performance of the samara-inspired UAV. A detailed energy efficiency analysis was conducted to understand the UAV's power requirements under various flight conditions. Energy consumption was monitored across different rotational speeds, altitudes, and angles of attack, revealing that higher rotational speeds and increased facing angles lead to significant energy demands due to higher lift and drag forces. Comparisons were made between powered flight and autorotation modes, with findings indicating that autorotation consumes up to 40% less energy, highlighting it as an effective energy-saving mode for descent. To further enhance efficiency, potential energy-saving measures were explored, including adaptive power modulation, which adjusts motor output based on real-time lift requirements, and optimized flight path planning to minimize high-drag maneuvers. These measures have the potential to extend flight duration, especially in low-power, hovering tasks. Compared to traditional UAV designs, the samara-inspired model demonstrated superior stability during vertical descent, with the spinning wing generating sufficient lift to support a payload of up to 200 g. The UAV's aerodynamic efficiency was also significantly higher than that of rotary-wing UAVs, particularly during low-speed flight.

To illustrate the practical value of the design, several potential application scenarios were analyzed in detail. For search and rescue operations, the UAV's stable autorotation and precise control make it ideal for hovering over specific areas, assisting in locating individuals in challenging terrains. Its ability to maintain stability in mild turbulence ensures reliability in diverse weather conditions. In precision agriculture, the UAV's low-energy consumption and extended flight time allow for effective crop monitoring and targeted pesticide or fertilizer application over large fields, increasing agricultural efficiency. These application scenarios highlight the UAV's adaptability and utility in real-world situations, showcasing how its bio-inspired design can meet the specific demands of tasks requiring prolonged flight, stability, and low-energy operation.

Overall, the samara-inspired design offers a promising alternative to traditional UAV configurations, particularly for applications that require stability and extended flight times. Future research will focus on improving the UAV's control algorithms and exploring additional use cases, such as search-and-rescue missions and precision agriculture.

6. Conclusion

This paper presented the design and aerodynamic analysis of a samara-inspired UAV, leveraging the natural stability and lift-generating characteristics of the samara seed to achieve stable spin flight. The UAV's aerodynamic performance was thoroughly evaluated using CFD simulations, which confirmed its ability to carry a payload of 200 g while maintaining stability and control.

The results of this study highlight the potential of samara-inspired designs for a wide range of UAV applications, particularly those that require precise control and extended flight times. However, further research is needed to optimize the UAV's energy efficiency and explore its scalability for swarm applications. Future work will focus on refining the control system and integrating more advanced sensor technologies to enhance the UAV's capabilities.

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